Exploring empathy in artificial intelligence: synthesis and paths for future research

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Abstract

Purpose – The current research elucidates the role of empathy in design of artificial intelligence (AI) systems in healthcare context, through a structured literature review, analysis and synthesis of academic literature published between 1990 and 2024.

Design/methodology/approach – This study aims to advance the domain of empathy in AI by adopting theory constructs context method approach using the PRISMA 2020 framework.

Findings – The study presents a current state-of-the-art literature to review the connections between empathy and AI and identifying four clusters showing the emerging trajectories in the field of AI and empathy in healthcare setting.

Originality/value – Despite a rise in empirical research, the potential pathways enhancing AI accountability by incorporation of empathy is unclear. The research aims to contribute to the existing literature on AI and empathy in the healthcare sector by carving out four distinct clusters depicting the future research avenues.

Keywords TCCM, Artificial intelligence, Empathy, Systematic literature review

Paper type Literature review

1. Introduction

For more than 250 years, disruptive innovations have remained the cornerstones of economic growth (Brynjolfsson and Mcafee, 2017). Especially, the advent of the fourth industrial revolution or Industry 4.0 technologies has accelerated global industrial growth (Morris *et al.*, 2018). Among these technologies, artificial intelligence (AI) is transforming the way we live, work and do business (Ho *et al.*, 2020; Thenral and Annamalai, 2021; Tutun *et al.*, 2023). With time, AI has significantly powered almost every routine of the activities, ranging from social media feeds to the online shopping experience (Kumar *et al.*, 2019).

AI depicts to a general idea that computers, by means of software and algorithm's assistance, can judge and execute tasks like individuals (Kumar *et al.*, 2019). AI has played a prominent role in elevating the standards of mankind. It can be used to train machines based on pre-defined grammatical sets with independent decision-making competencies, eliminating the demand for a human presence (Spennemann, 2007). However, with these technological advancements, the concerns regarding the moral judgments of AI, i.e. empathy, remain unanswered (Floridi *et al.*, 2018; Müller, 2020). AI is not sophisticated sufficiently to comprehend cultural disparities in expressing and interpreting emotions, making it difficult to draw accurate inferences (Purdy *et al.*, 2019).

A thorough exploration is needed to comprehend whether AI can assist in empathy and compassion, which are essential to delivering patient-centred care (Bærøe *et al.*, 2020a; Kerasidou, 2020). The patients disturbed with mental conditions face challenges getting professional assistance because of social stigma and reluctance (Sussman *et al.*, 2022). In addition, nations are facing a scarcity of mental health experts. AI can also be instrumental in cost-cutting measures in healthcare sector, estimated to minimize yearly US healthcare costs by US$150bn in 2026 (Bohr and Memarzadeh, 2020). As stated by Forbes, the most significant application areas for AI would be supervisory workflows, image interpretation, robotic surgery, virtual assistants and clinical decision assistance (Jimma, 2023). Especially for patients dealing with mental issues, AI can furnish monitoring and counselling services by enhancing their natural language processing competencies or emotion detecting algorithms (Bohr and Memarzadeh, 2020).

These challenges have not been addressed in the current empathic AI literature. For this objective, a review that integrates past literature on this area and draw meaningful insights is imperative. The review aims to find significant factors instrumental for an empathetic AI by conducting a systematic literature review (SLR) of 42 research articles published in the field, extracted from the Scopus and Web of Science (WOS) database and after implementing inclusion-exclusion criteria. The study addresses the following research objective (RO):

> *Research Objective 1:* To examine worldwide advancements in the scholarly understanding of the relationship between artificial intelligence (AI) and empathy within the healthcare industry.

To identify the progression in the field of empathy and AI, the present research has used the theory constructs context method

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(TCCM) framework, using Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 statement (Sarkis-Onofre *et al.*, 2021; Weinfurt, 2024). TCCM is a commonly used methodological approach for examining the evolution of a particular research field (Paul and Rosado-Serrano, 2019; Paul and Feliciano-Cestero, 2021; Roy Bhattacharjee *et al.*, 2022). By highlighting the emerging trajectories in the field, the present research also offers future research avenues in this field.

The current research offers significant contributions. The present research is a notable study that does a SLR on the topic of AI and empathy, according to the standards outlined in the PRISMA 2020 guidelines. The current research offers a unique set of theoretical and practical implications for practitioners and academicians by taking a blend of empathy and AI. The findings augment the existing health technology literature by emphasizing on empathic emotions and algorithm aversion by Årnfelt (2021), with a focus on real-life experiences of medical practitioners and their views on usage of empathy-driven AI. Regarding practitioners, current research suggests testing instances wherein empathy driven AI has been able to meet up the ongoing demand for medical practitioners, exhibiting improved health results. The present research is also crucial about addressing the current challenges faced by mental healthcare professionals and coming up with significant propositions. The research provides a definitive path to tackle the psychological difficulties faced by individuals worldwide, particularly during periods of uncertainty and beyond, such as the post-COVID-19 era. During this time, individuals are striving to maintain a balance between their work and personal lives in a hybrid culture. This challenge can be addressed with the help of AI that is driven by empathy.

The rest of the paper is structured as, firstly, a comprehensive rationale behind the applicability of the SLR methodology of 42 journal articles for this study is introduced. After that, both the descriptive and the thematic analysis of outcomes that have exhibited the base for proposing an integrative framework of AI and empathy have been presented. In conclusion, opportunities and suggestions for future research in this area have been discussed along with the final considerations.

2. Conceptual boundaries of the review

Prior scholars from psychology and organizational science have conceptualized empathy as a complex construct comprising of a range of experiences and functions used in several ways by therapists and psychologists of diverse orientations for distinct objectives. For this review, we adopted the definition given by Barrett-Lennard (1981), which classifies empathy in three subcategories: one being therapist's experience ("empathic resonance"), second, being observers' perspective ("expressed empathy") and the last being the client's experience ("received empathy"), structured in a sequential process model (Barrett-Lennard, 1981). In another study, empathy was identified as one of three characteristics of a therapist, together with warmth and authenticity, that provide the necessary and adequate atmosphere for therapeutic growth to occur (Rogers, 1957). So, empathy can be illustrated as the ability to recreate another person's perspective to experience the world from his or her point of view.

In healthcare context, empathy is defined by as "the competence of health care professionals to precisely comprehend patients, emotionally and cognitively, as if they're in the patient's shoes, however without compromising their position" (Rogers, 1957). Empathetic communication is a teachable, learnable ability that has practical advantages for both clinicians and patients, according to researchers and education professionals. Competent empathetic communication strengthens the therapeutic effectiveness of physician–patient engagement (Ioannidou and Konstantikaki, 2008). Empathy has often been linked with emotional intelligence, which is a significant factor in comprehending the patient's viewpoint, for a healthcare professional (Ioannidou and Konstantikaki, 2008).

Emotional intelligence is an individual's capability:

- to comprehend his/her feelings;
- to heed to others and to sense them; and
- to convey his/her emotions in a constructive manner (Goleman, 1998).

Furthermore, "emotional intelligence" is a concept involving perception, expression and restrain of emotions, self-control, empathy and several other prominent constituents (Ioannidou and Konstantikaki, 2008).

Artificial intelligence, on the other hand, has been portrayed as the knowledge and engineering of building intelligent robots, primarily intelligent computer processes (McCarthy, 2007). AI illustrates a chance to create an economically feasible option for an expensive product or service. The approach AI takes in this regard is generating information that was not available before (Agrawal *et al.*, 2017). AI can assist in three prominent business requirements: automating firm proceedings, attaining insight through data scrutiny and appealing to consumers and personnel (Ronanki and Davenport, 2018). Generative AI, a specific branch of AI, is projected to grow into a market worth $1.3tn by the year 2032 (Bloomberg, 2023).

Prior studies have noted the significance of AI in healthcare, with an emphasis that its usage would result in significant advancements in all aspects of healthcare, from diagnosis to therapy (Bohr and Memarzadeh, 2020). AI has been considered a powerful instrument, bringing interest into the purpose and value of empathy and compassion in health care (Kerasidou, 2020). In patient-focused care, empathy enables healthcare professionals to comprehend the patients' perspective, thus assisting health professionals in adapting care to foster the patients' values and address their individual requirements (Kerasidou, 2020). Researchers have also focused on a concept, known as "humanization of robotic systems", (AI), which is an attempt to construct robots that more closely resemble human look and behaviour, such as the exhibition of humanlike cognitive and emotional states. This can be accomplished by employing social (e.g. communication, non-verbal behaviour, personality, emotions and empathy), ethical (e.g. morality and ideals) and religious (e.g. faith, culture and tradition) competencies (Hashim *et al.*, 2017).

AI systems should be designed to enable value-plurality, signifying the potential of diverse patients holding distinct values with distinct preferences concerning their care (Goodman *et al.*, 2020; Kerasidou, 2020). Thus, the current study addresses this concern by coming up with a framework that would guide healthcare practitioners to use an empathetic artificial intelligent system. By presenting a SLR of empathy implication of AI, its antecedents and outcomes, the current research is directed at (i) offering a thorough insight of the central principles and the true

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significance of the concept, (ii) strengthening the current understanding of the conceptualization of empathy beyond the emotional intelligence context and (iii) propounding a theoretical framework that presents empathy as a significant feature of AI valuable for the intellectually disabled patients, and correspondingly, the healthcare sector. On these grounds, the current research provides a base for proposing effective techniques for refining the position of mental health standards around the globe.

3. Methodology

This research incorporates the SLR approach, according to the PRISMA 2020 principles (Selcuk, 2019; Page *et al.*, 2021; Sarkis-Onofre *et al.*, 2021). These principles provide a comprehensive set of evidence-based elements for reporting in systematic reviews and meta-analyses. An SLR is a meticulous method of structuring, presenting and summarizing pre-existing research papers. The objective of this is to assist readers in obtaining a thorough understanding of the present level of knowledge in a certain area of study, as well as anticipated future advancements. The investigation conducted found papers that elucidate the significance of empathy in the construction of AI systems specifically tailored for healthcare applications. To investigate the study topics, an SLR of existing literature on this subject has been conducted (Kitchenham, 2004; Brereton *et al.*, 2007). The systematic method was chosen for its capacity to provide assessments of excellent quality and clear, consistent examination of the current literature (Dybå and Dingsøyr, 2008). By providing rigorous, transparent and repeatable search results, it allows academics to assess, evaluate and incorporate existing literature on a certain topic. This helps to identify the current knowledge, research gaps and recommendations for future investigations (Denyer and Tranfield, 2009; Papathanassoglou *et al.*, 2017). The objective of current research is to provide specific strategies and actions that healthcare organizations might use to expand their influence. Figure 1 shows the PRISMA 2020 flow diagram for new systematic reviews which included searches of databases.

3.1 Literature search and assessment

3.1.1 Search string strategy

This study used a methodical approach based on the PRISMA guidelines (Page *et al.*, 2021; Sarkis-Onofre *et al.*, 2021). The article's title and abstract followed established guidelines, and the review objectives were clearly outlined in the introduction, which also specified the primary criteria for including and excluding items. The search string was constructed based on the research questions and included a compilation of synonyms, acronyms and alternative spellings. Due to the diverse nature of AI and its numerous categories, subcategories and methodologies, it was crucial to develop an exhaustive strategy. Therefore, a search strategy was implemented that was in accordance with the PRISMA recommendations. The search keywords were divided into two clusters: Group A keywords focused on AI, i.e. 'Artificial Intelligence', 'Machine Intelligence', 'Neural Networks', 'Robotics' and 'Automatic Technology', and Group B keywords constituted terms linked to empathy, i.e. 'Empathy', 'Compassion', 'Affinity', 'Congeniality' and 'Sensitivity'. Scopus and WOS databases have been used to retrieve pertinent studies. Table 1 displays the papers that were obtained by using various keywords related to AI and empathy from different databases.

For assortment of relevant articles, two members of the research team conducted an independent search for articles, contrasted the results and reached the final list of 42 articles (Kaur *et al.*, 2021). In order to measure the progression in AI-empathy research over time, the research articles were sorted as conceptual or empirical (MacInnis, 2004; Yadav, 2010; Roy Bhattacharjee *et al.*, 2022). To evaluate the inter-rater reliability, we evaluated Cohen's kappa, which was found to be 0.88 (Cohen, 1960). A defined inclusion/exclusion criterion has been adopted for deriving the final sample of studies for review.

3.1.2 Inclusion and exclusion criterion

Due to the limited number of research papers in this domain, it is crucial to determine the specific criteria for inclusion while still adhering to the recommendations outlined in the PRISMA document (Page *et al.*, 2021; Sarkis-Onofre *et al.*, 2021). The journal articles were chosen from the period of 1990–2024 to include all research publications that specifically addressed these subjects throughout the research. Every article has been shortlisted with the following terms in their title, abstract or keywords: Search terms "Business, Management, and Accounting" (in Scopus) and "Business", "Management", "Public Health", "Psychiatry" and "Social Sciences, Interdisciplinary" (in WOS). The journal articles were selected from 1990 to 2024 to include every possible research article that addressed these themes in their explorations. An aggregate of the standards referred in the previous step in context to the time span and journal selection was implemented to procure the primary list of 76 articles. Secondly, the inclusion list comprised of peer-reviewed journals, journal selection criteria included taking articles from A*, A, B and C category journals of the Australian Business Deans Council (ABDC) journal categorization, articles in English language were included, subject area incorporated AI-empathy articles from the business domain, and articles had to be a full paper.

The ABDC's journal quality list has been selected as the foundation for the study due to its perceived relevance within the business academic community (Moosa, 2016; Hair *et al.*, 2019; Jaafar *et al.*, 2021). ABDC's ranking approach includes research impact, in accordance with other global rankings and measures (de Villiers and Hsiao, 2018). The ABDC list provides a ranking of journals in all fields of business and economics (Hair *et al.*, 2019). The journals are classified into four grades: A*, A, B and C (Hair *et al.*, 2019; Hirschberg and Lye, 2020). The ABDC website has recorded a sequence of public conversations about these measures, which have been suggested for assessing research inside an institution and across many universities (Hirschberg and Lye, 2020). These are the justifications for choosing the ABDC list of categorizations.

Similarly, the exclusion criteria involved eliminating duplicate records, non-contextual research articles, papers from non-paired records, along with excluding grey literature (consisting of doctoral thesis, conference articles and workshops). It is a regular practice to have articles comprising 20, 25 or 30 years of studies in review papers (Furrer *et al.*, 2008). In addition, it is advisable that a bare minimum of 10 years to be included for a SLR (Rialp *et al.*, 2005; Paul and Feliciano-Cestero, 2021).

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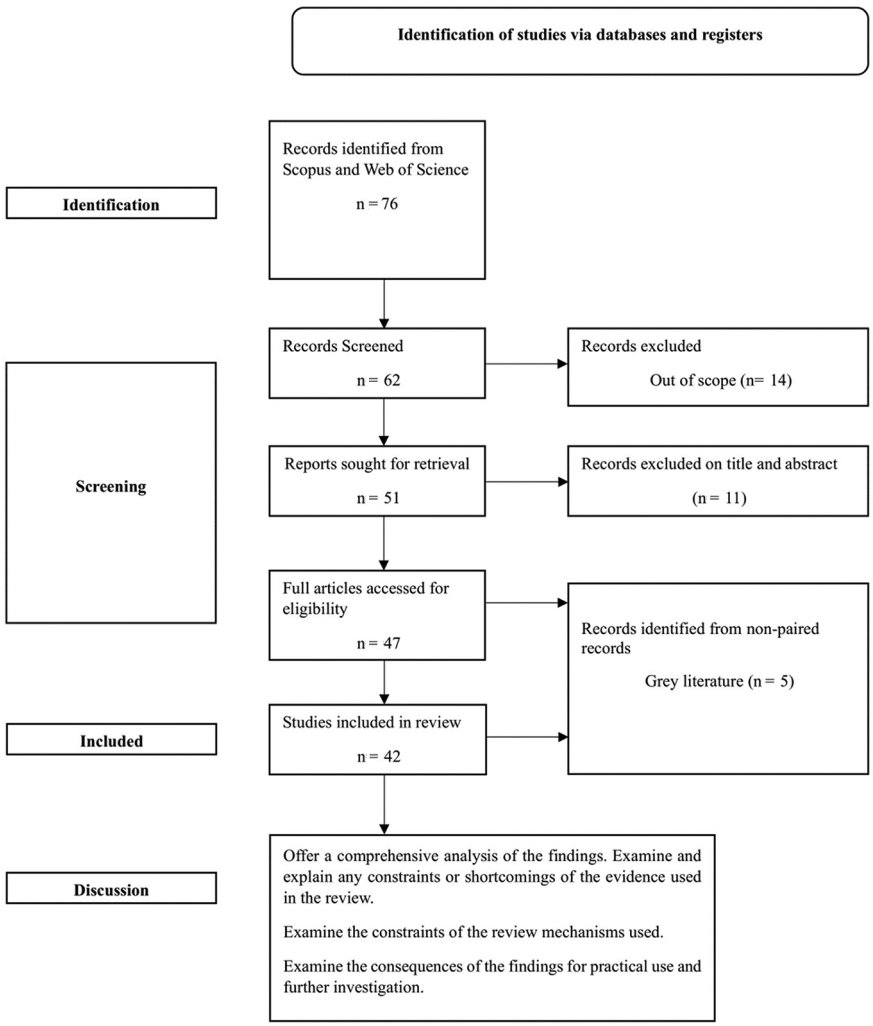
Figure 1 PRISMA 2020 flow diagram for new systematic reviews which included searches of databases



Source: Figure by author

Table 1 Extracted articles by using several keywords for AI and empathy in Scopus and Web of Science

Group A (Keywords used for artificial intelligence)	Operator	Group E (Keywords used for empathy)	Extracted articles
Artificial Intelligence	AND	Empathy	24
Artificial Intelligence	AND	Compassion	21
Machine Intelligence	AND	Affinity	16
Neural Networks	AND	Congeniality	Nil
Robotics	AND	Sensitivity	15
Automatic Technology	AND	Empathy	Nil

Source: Table by author

The current research deploys several inclusion and exclusion criteria's for compressing the research articles published in the domain of AI and empathy. The inclusion standards included selecting articles published between the period 1990 and 2024, selection of articles from A*, A, B and C category journals, articles written in English, subject area including articles from Health Sciences: All Health Sciences and Social Sciences: Business, Management and Accounting, and articles being a full paper.

Regarding the exclusion criteria, the articles had to be screened by eliminating duplicate records, non-contextual

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articles and articles from non-paired records. The exclusion also included removal of articles from grey literature (consisting of doctoral thesis, conference articles and workshops).

3.2 Article selection and data extraction

In the next step, a screening procedure was followed based on PRISMA guidelines, which reviewed the articles based on title, abstract and keywords, leading to 76 articles (Page *et al.*, 2021; Sarkis-Onofre *et al.*, 2021; Weinfurt, 2024). This was then followed by a deep screening process under which the thorough reading of research articles was performed, and articles out of the scope were removed. This led to the elimination of fourteen articles. The remaining 62 articles were carefully read to eliminate non-contextual research articles, leading to 51 articles. In the subsequent steps, articles from non-paired records, as well as grey literature were removed, resulting in the final 42 articles to be assessed by the systematic review approach. Figure 2 presents an overview of the stepwise article selection process exerted for the study.

4. Results and analysis

4.1 Theory constructs context method analysis

SLR method based on the TCCM review framework was used in the study to furnish a holistic view of the exploration. In contrast to other review approaches, the TCCM framework is most used through diverse research areas, and academicians have indicated it to be the most competent tool for literature review TCCM analysis facilitates in bridging the gaps identified in the previous studies and offers pathways for further scope (Sharma *et al.*, 2020; Srivastava *et al.*, 2020; Rajan *et al.*, 2021). The review articles using the TCCM framework were printed in impactful journals comprising *Journal of Business Research, International Marketing Review* and *International Business Review* (Srivastava *et al.*, 2020; Chaturvedi and Dasgupta, 2023). The framework is used to scrutinize the progression of the area over the past three decades and present future research directions.

The 5W1H is an interrogative strategy and a method for resolving problems that addresses all the fundamental components of an issue, namely, what, who, when, where, why and how (Jia *et al.*, 2016; Herjanto *et al.*, 2022; Paul *et al.*, 2023). When compared to other SLR approaches like the 5W+1H framework, which emphasizes openness and systematic reporting, TCCM provides more comprehensive analysis and theoretical integration, making it a stronger tool for systematic review. Moreover, the methodical and systematic character of TCCM improves its usefulness and adaptability in other fields. The systematic classification of data facilitates the identification of study gaps, trends and future directions. The framework's focus on methodological intricacies and contextual depth enhances the analytical rigor and usefulness of the review.

Compared to other SLR approaches such as the 5W+1H framework, also known as the 6W framework proposed by Callahan (2014), which focuses on openness and systematic reporting, TCCM offers a more thorough analysis and theoretical integration, making it a more powerful tool for systematic review (Paul et al., 2021, 2023; Roy Bhattacharjee *et al.*, 2022). The 5W+1H review includes a constraint that it must be used with another framework, such as the TCCM or antecedents-decisions-outcomes (ADO) framework, to evaluate literature (Paul *et al.*, 2023). The ADO framework, as presented by Paul and Benito (2018), lacks the necessary explanations of theories, settings, and methodologies that are essential for guiding future research. The input-mediator-outcome (IMO) paradigm, as proposed by Ilgen *et al.* (2005) and Mathieu *et al.* (2008), has mostly been used to analyse team processes. However, it has the potential to be adapted to many other organizational processes. Therefore, the TCCM framework provides a comprehensive overview of a study topic, enabling a critical assessment of its theoretical foundations, methodological strategies, features and contextual factors (Aaltonen, 2020; Sharma *et al.*, 2020; Paul *et al.*, 2021, 2023). While frameworks like as ADO, IMO and ADO-TCM have their own advantages, they are ineffective to replicate the comprehensive range and smooth integration provided by TCCM (Paul *et al.*, 2023). The widespread acceptance and use of TCCM in respected publications provide proof of its effectiveness and reliability, confirming it as the most appropriate choice for conducting literature reviews.

4.1.1 Theories

Table 2 explains the various theories that have been used in the context of empathy and AI. The affective or cognitive approach of empathy has been used by Leite *et al.* (2013a). Empathy has been illustrated as an emotional response that consequences from one of the following empathic arousal processes: classical preparation, the direct alliance of prompts from others and mediated affiliation (Hoffman, 2001).

The notions of empathy, compassion and trust have been used by Kretzschmar *et al.* (2019), D'Alfonso (2020) and Balcombe and de Leo (2021). The technological singularity (TS) has been implemented by Shuaib *et al.* (2020). The TS is a potential point in the future when AI is projected to surpass human intellect (Solez *et al.*, 2013; Bavaresco *et al.*, 2020). In a strict sense, TS in health care would entail those medical practitioners will be replaced by AI-enabled robots and peripheral devices (Smolin, 2020). Ortony–Clore–Collins (OCC) theory was used by Paiva *et al.* (2017). OCC characterize fortunes-of-other feelings as empathic emotions and incorporate several local factors that determine the strength of an empathetic emotion in their cognitive appraisal theory of emotion (Clore and Ortony, 1988). Table 2 contains a quick summary of the list of theories that have been found in the peer-reviewed literature.

4.1.2 Context

With reference to the context, the current research has examined the research institutions, countries, scholars and area of research that have been explored concerning the AI-empathy domain. They have been briefly discussed in Table 4, along with the year of research conducted. Most of the research has been conducted in the USA (six). The research areas have broadly explored the contemporary evolutions in healthcare AI and suggestions for medical practitioners (Jiang *et al.*, 2017; Maruthappu, 2018; Matheny *et al.*, 2018; Goodman *et al.*, 2020). Another research has also broadly discussed about TS, which illustrates the possible future wherein AI will surpass human capabilities (Shuaib *et al.*, 2020). England comes second in the exploration of empathy-driven AI, with four articles. The diverse research conducted in the area focusses on the implications of AI-chatbot in mental health assistance, the

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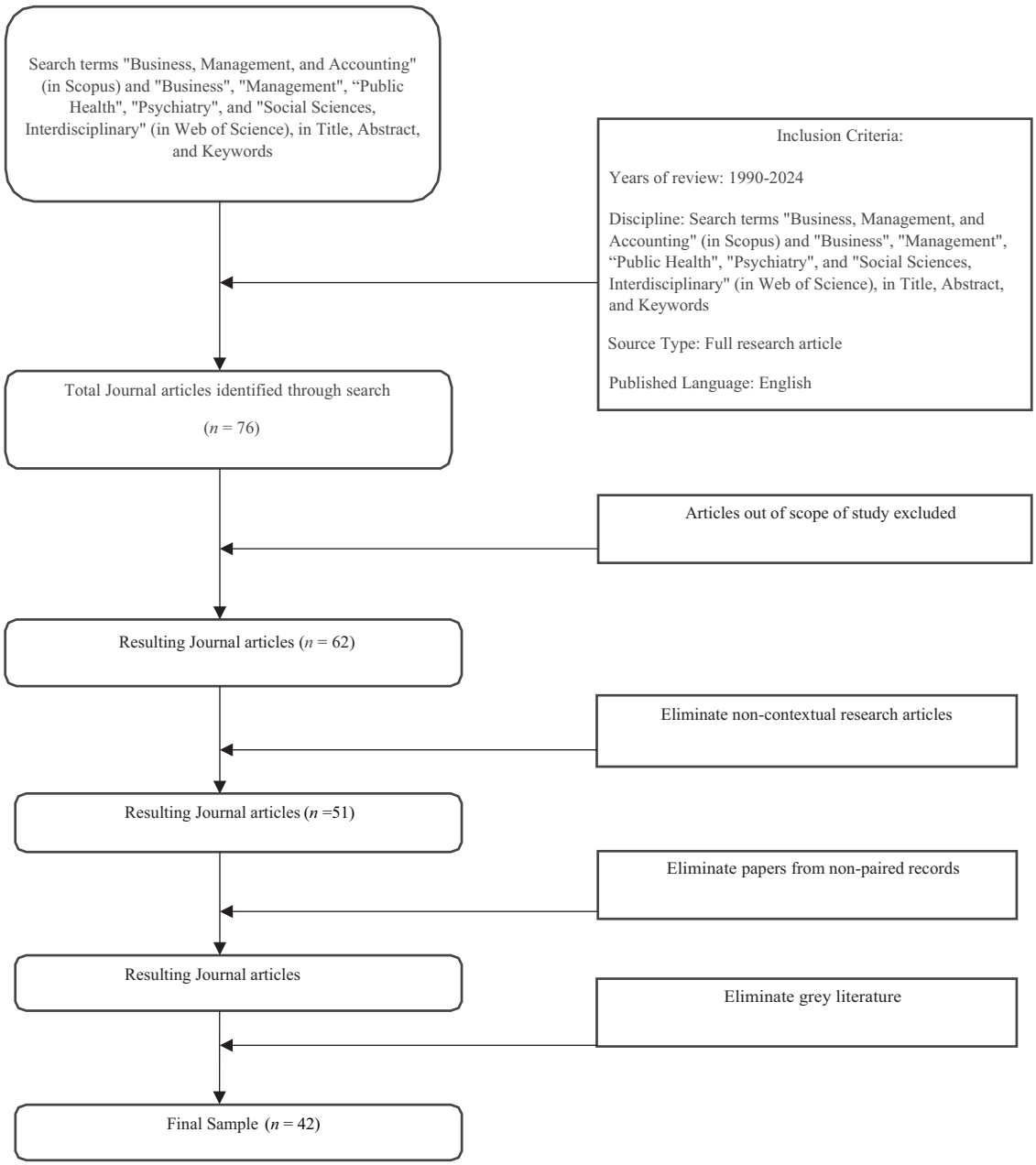
Figure 2 Article selection process for the structured review



Source: Figure by author

significance of emotional intelligence in an empathic AI-chatbot, along with a focus on the set of AI guidelines for directing firms to take responsible AI operations (Kretzschmar *et al.*, 2019; Devaram, 2020; Kerasidou, 2020; Liu *et al.*, 2021).

Australia, with three publications on the area, has illustrated the outcomes of AI healthcare research over a year, emphasising the approaches through which AI has been deployed across mental health firms. Premier institutions like Monash University, University of Melbourne and Griffith University have portrayed that AI, will indeed become a significant tool in medicinal practices around the globe (Loh, 2018; D'Alfonso, 2020; Balcombe and de Leo, 2021).

Switzerland and Sweden, from the European continent, have published two studies, each. The studies have carried out by World Health Organization (WHO) from Australia and KTH Royal Institute of Technology along with Linköping University, from Sweden, respectively. The research outcomes from Sweden have discussed the aspects of an autonomous robot with empathic competencies behaving as a companion to individuals (Leite *et al.*, 2014; Leite, 2015).

Another research has emphasized on the algorithm aversion of AI, which highlights on the fact that individuals tend to give more emphasis on the recommendations given by a human, rather than an algorithm. In other words, people do not accept the

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Table 2 Notable theories used in AI-empathy research

Authors	Theory used in the study	Description of theory
Leite, (2015) Leite *et al.* (2014)	Cognitive empathy	Cognitive empathy, or empathic accuracy, refers to the ability to possess comprehensive and precise understanding of another individual's thoughts and emotions
Leite *et al.* (2014) Paiva *et al.* (2017)	Ortony–Clore–Collins theory	The idea proposed by Ortony, Clore and Collins posits that emotions arise because of specific cognitions and perceptions. Thus, it focuses only on the cognitive factors that trigger emotions
	Algorithm aversion theory	Algorithm aversion refers to the tendency of people to dislike algorithms, even though algorithms frequently outperform humans
Devaram (2020)	Emotional intelligence	Emotional intelligence (EI) refers to a person's ability to responsibly manage one's own emotions and comprehend the emotions of others. The five essential components of EI are self-awareness, self-regulation, motivation, empathy and social skills
Shuaib *et al.* (2020) Solez *et al.* (2013)	Technological singularity (TS)	The technological singularity refers to a theoretical future moment when the rate of technology advancement becomes unstoppable and irreversible, leading to unpredictable outcomes for human society
	Algorithmic appreciation	Algorithm appreciation refers to an individual's inclination or propensity to depend on algorithms while making decisions

Source: Table by author

evidence-based algorithms, which precisely forecast the future than human forecasters (Årnfelt, 2021). The WHO reports, on the other hand, have addressed the assessment of engagement an AI-enabled, mental well-being app, Wysa, on users with depression. The report outcomes displayed that it does significantly improves the status of individuals with depression, and it has a promising future (Inkster *et al.*, 2018; Woodward *et al.*, 2020). Another report by WHO has emphasized on the ethics and managing AI, which is imperative for an organization's sustenance and growth (World Health Organization, 2023). The ethical significance of AI in healthcare has also been briefly outlined by research conducted in Norway that discusses the conflicts arising due to this concern (Bærøe *et al.*, 2020b). Table 3 highlights the backgrounds of research conducted on the area.

4.1.3 Characteristics

In the current research, a companion's effective state, perspective taking and empathetic responses have been observed to be significant factors which assist in explaining AI and empathy research (Leite *et al.*, 2013b; Leite, 2015). Thereafter, mental health has been spoken in one of the research projects as a state of well-being in which a person recognizes his or her own potential, is able to cope with difficult situations, serves its purpose and contributes to society (D'Alfonso, 2020).

In a subsequent research, trust have been highlighted as one of the significant constituents in building in a relationship with the empathy aided AI and the patients who are suffering from mental disorders (Kerasidou, 2020; Kraft, 2020). The various explorations conducted by various scholars across the globe have emphasized on the consumer trust as one of the bases through which the empathy aided AI can be accepted in the healthcare sector, with a specific focus to the medical practitioners using it as one of the mediums for their clinical activities (Liu *et al.*, 2021).

Other research projects have also emphasized on the relevance of this type of AI and how it can pave a new direction in the area relating to treatment of mental health concerns (Vasudevan *et al.*, 2021). Table 4 has portrayed the diverse variables employed by several studies concerning the empathy and AI literature. The table also displays the sources along with description for each variable for a clear understanding in the progression of research conducted in the domain.

4.1.4 Methodologies

In context to methodologies, most articles from the period of 1990–2023 comprised of conceptual research articles that is, 18 articles out of the 42 articles chosen for the systematic review. Only five research articles among the 42 articles were empirical in nature and used quantitative research techniques to come to the conclusion. There are five empirical studies which focused on several empirical measures. One of them focused on analysis of 31 respondents, doing the analysis quantitatively and qualitatively, and using a questionnaire concerning interpersonal characteristics (Leite *et al.*, 2013a; Leite, 2015). Some of them consisted of measuring friendship functions using a questionnaire; using Wysa application, which is a software which allows you to converse with a penguin chatbot about everything that is on your mind (Leite *et al.*, 2013b). Wysa uses text messaging and self-reported depressive symptoms using the Patient Health Questionnaire (Leite *et al.*, 2013a; Inkster *et al.*, 2018).

The conceptual articles however, included exploration on the empathetic robots based on the theory and processes of empathy observed in humankind (Paiva *et al.*, 2017); summarizing the last twelve months of healthcare research in AI over several clinical specialties, and to highlight the present capabilities and deficiencies, as well as challenges, associated with this new innovation (Loh, 2018); exploring the competencies of three chatbots for mental health assistance (Kretzschmar *et al.*, 2019); concentrating on three of the most common approaches that AI is often used in psychological health: natural language processing of medical writings and social networking information, personalized sensing or digital behavioural genetics and AI chatbots (D'Alfonso, 2020); using a case based method based on global health care sector for attaining a trustworthy AI (Bærøe *et al.*, 2020b); applying a sentimental analysis based on video-supported emotion identification and voice-based emotion identification (Devaram, 2020); designing a scoping review article of the AI chatbots in healthcare psychology and psychotherapy (Bendig *et al.*, 2019); and conducting an in-depth interviews with 25 medical practitioners (Liu *et al.*, 2021).

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Table 3 Background of research conducted on the area

Geography of research	Authors and year of research	Institution	Area of research
Sweden	Leite *et al.* (2013)	KTH Royal Institute of Technology	Autonomous robot with empathic capabilities behaving as a social companion
	Årnfelt (2021)	Linköping University	Algorithm aversion of AI
Portugal	Paiva *et al.* (2017)	University of Lisbon	Computational empathy
		Research Centre in Digitalization and Intelligent Robotics (CeDRI)	Employing digital automations, such as artificial intelligence (AI) in the advancement of ingenious rehabilitation capabilities primarily on mental health realignment
Australia	Loh (2018)	Monash University	Outline past 12 months of health research in AI, across different medical specialties
	D'Alfonso (2020)	University of Melbourne	Emphasize on three major ways that AI is being employed in mental health
	Balcombe and De Leo (2021)	Griffith University	Digital mental health
Switzerland	Inkster *et al.* (2018)	World Health Organization	Evaluation of engagement levels of an AI-enabled, conversational mobile mental well-being app, Wysa, on users with depression
	Guideline		Ethics and control of artificial intelligence
England	Kretzschmar *et al.* (2019)	University of Oxford	Chatbots in mental health support, e-mental health
	Bærøe *et al.* (2020)		Empathy, compassion and trust in healthcare
	Devaram (2020)	Bournemouth University	Emotional intelligence in empathic chatbot
	Liu *et al.* (2021)	Newcastle University	Construct a set of responsible AI principles for guiding organizational accountable AI operations
USA	Jiang *et al.* (2017)	University of California, San Francisco	Recent developments in healthcare AI and their implications for general practitioners
	Buch, *et al.* (2018)	Massachusetts General Hospital	Healthcare implications of AI and future scope
	Matheny *et al.* (2018)	National Academy of Medicine	Assist unification of AI into health care workflow
	Goodman *et al.* (2020)	University of Arizona	Artificial intelligence in the health industry
	Kraft (2020)	University of Washington	Relational significations of machine learning in healthcare
	Shuaib *et al.* (2020)	UT Southwestern Medical Center	Technological singularity (TS), artificial intelligence (AI) in healthcare
Canada		College of Family Physicians Canada	Need for human connection in observation and data interpretation (along with AI assistance)
Qatar	Abd-alrazaq *et al.* (2019)	Hamad Bin Khalifa University	A synopsis of attributes of chatbots used by people for their mental health as stated in the empirical literature
Norway	Bærøe *et al.* (2020b)	University of Bergen	Ethical conflicts of artificial intelligence systems in healthcare
South Korea	Oh *et al.* (2019)	Soonchunhyang University	Examine awareness of AI among Korean doctors, with their attitudes toward the medical practice of AI
India	Chakriswaran *et al.* (2017)	Vellore Institute of Technology	To review a few of the key researches that have been conducted far and to present an insight of sentimental analysis models in the domain of emotional AI-driven sentimental analysis
	Vasudevan *et al.* (2021)	Christ University, Bangalore	AI implications on mental health issues
China	Shi *et al.* (2021)	Dalian University of Technology	Exercise of automatically detected metaphors in text to forecast mental health concerns

Source: Table by author

5. Research clusters or potential domains of research

To add to the field of empathy and AI in healthcare, the present study classifies the 43 research publications into four major groupings. Furthermore, the study aimed to identify dominating constructs and developing future themes within each cluster, both of which have the potential to significantly contribute to the domain. To accomplish this purpose, a thorough examination of the 43 research papers was conducted with the intention of grouping the articles into four unique clusters. Cluster 1 has five articles, Cluster 2 has six, Cluster 3 has 16 and Cluster 4 has 12. Each cluster proposes novel avenues of investigation for experts engaged in the topic. Figure 3 depicts the same approach.

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Table 4 Variables identified from the AI-empathy literature

Variables identified	Sources	Description
Companion's affective state, perspective-taking, empathic responses	Leite *et al.* (2013)	Affective state is an experience of sensing the underlying emotional state. Perspective refers to the capacity to go beyond your own point of view in order to reflect how someone else might think or feel about something. Empathy is the ability to comprehend another person's experience, express and confirm that awareness with that person, and then act in a sensible manner
Mental health	D'Alfonso (2020)	Mental health is a stage of well-being in which a person recognizes his or her own potential, is able to cope with stressful situations, can work efficiently, and can contribute to his or her society
Trust	Kerasidou (2020)	Trust is the characteristic of believing in the integrity and dependability of individuals
Respect, trustworthiness	Kraft (2020)	Respect is a feeling of appreciation or regard. Trustworthiness is the attribute of an individual or object that fosters trust
Emotional intelligence, mental health well-being	Devaram (2020)	"Emotional intelligence" is a concept involving perception, expression and restrain of emotions, self-control, empathy, and several other prominent constituents (Ioannidou and Konstantikaki, 2008). Mental wellbeing indicates well-being on all levels: emotional, cognitive and interpersonal
Consumer trust	Liu *et al.* (2021)	Consumer trust is the trust that firms will do what they promise they will and would not intentionally affect customers
Mental health issues	Vasudevan *et al.* (2021)	Mental health issues relate to a broad scope of disorders that influence emotions, thought and personality

Source: Table by author

5.1 Cluster 1: Social robots and empathy

The first cluster discusses the research articles that explore the role of empathy in social robots. The prominent characteristics emerging in this cluster are voice characteristics (pitch), language cues (empathy/humour expression), complexity of the robot's social behaviour, pseudo-development mechanism, recognising and speaking languages among others (Castellano *et al.*, 2013; Niculescu *et al.*, 2013). The most cited research articles in the clusters are Leite *et al.* (2013b), total citations (TC) = 532; Niculescu *et al.* (2013), TC = 327 and Chakriswaran *et al.* (2017), TC = 77. In their study, Niculescu *et al.* (2013) investigated the impact of basic auditory and linguistic elements of spoken language, including voice pitch and language signals like empathy and humour expression, on the quality of contact using a social robot receptionist. Moreover, the article further demonstrates that empathy influenced how users perceived the social robot's responsive behaviour and the ease of interaction (Niculescu *et al.*, 2013). Empathic behaviour generation, affect recognition and emotional AI-driven sentiment analysis have been identified as crucial factors in enhancing empathy in social robots (Leite *et al.*, 2013b; Chakriswaran *et al.*, 2017). In regard to future research avenues, the cluster speaks about the need to make a substantial contribution in addressing sentiment analysis's issue of helping the machine comprehend intense human emotions (Chakriswaran *et al.*, 2017). Furthermore, unique assessment metrics and surveys validated specifically for empathetic agent research should be among the computational empathy community's primary areas of focus. Another essential factor for the future of computational empathy is to use cutting-edge research in emotional computing and user modelling (Yalçın and DiPaola, 2020).

5.2 Cluster 2: Empathy in human–robot relations

Cluster 2 discusses the significance of empathy in the relationships between humans and robots. The prominent characteristics in this cluster are companion's affective state, perspective-taking, empathic responses, affect recognition, human–robot interaction, contextual information among others (Leite *et al.*, 2013b; Castellano *et al.*, 2014, 2017). The most cited research articles in the clusters are Leite *et al.* (2013b), TC = 355; Paiva *et al.* (2017), TC = 321; and Leite *et al.* (2014), TC = 277. The clusters definition of empathy includes the cognitive capacity to comprehend and grasp the emotions and ideas of others by engaging in perspective-taking. This involves the ability to mentally place oneself in another person's position and envision how they could be experiencing (Leite *et al.*, 2013b). According to Paiva *et al.* (2017), empathy is a psychological process in which an individual experiences emotions that align more closely with another person's circumstance than with their own circumstances. The cluster states that cognitive and social sciences have not yet completely explored the psychological phenomena of empathy. Consequently, the empathetic behaviours and prosocial techniques used by empathic virtual agents and robots are context-dependent and challenging to apply universally across different domains (Castellano *et al.*, 2014, 2017). Therefore, based on these reasons, the cluster suggests investigating the degree to which humans perceive the support offered by the robot as authentic. In addition, further investigation should be conducted to examine the consequences of a sudden malfunction or the inability of the robot to exhibit appropriate supporting behaviour in response to a seemingly non-important scenario (Leite *et al.*, 2013a, 2014). Another relevant field of study is investigating emotional signals and empathetic feelings, which arise by means of the manipulation of affective relationships, mood and personality (Rodrigues *et al.*, 2012; Correia *et al.*, 2018).

5.3 Cluster 3: Conversational agents and mental well-being

Cluster 3 explores the research articles concentrated on conversational agents/chatbots and their role in influencing the mental well-being of patients/individuals. The prominent

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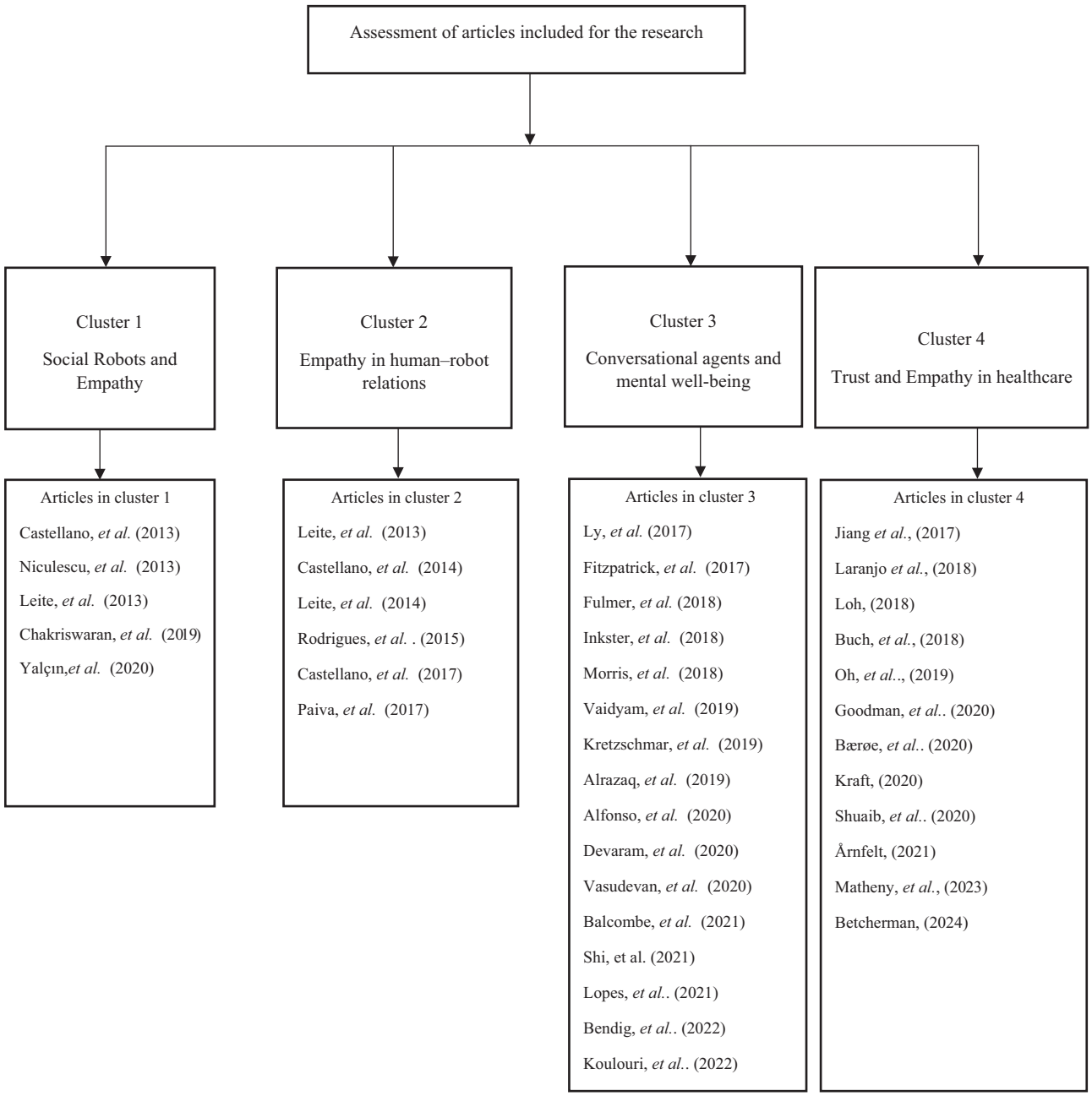
Figure 3 Prominent clusters identified from the literature



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characteristics in this cluster are task-orientation, dialogue management, dialogue initiative, conversational agents behaviour, digital mental health, innovations in mental health systems and user-centred design among others (Abd-alrazaq *et al.*, 2019; Kretzschmar *et al.*, 2019; Bendig *et al.*, 2022; Koulouri *et al.*, 2022). The most cited research articles in this cluster are Fitzpatrick *et al.* (2017), TC = 1783; Inkster *et al.* (2018), TC = 594; Fulmer *et al.* (2018), TC = 421; Abd-alrazaq *et al.* (2019), TC = 372 and Ly *et al.* (2017), TC = 332. Fitzpatrick *et al.* (2017) research aimed to assess the practicality, desirability, and initial effectiveness of a completely automated conversational agent in providing a self-help programme for undergraduates who acknowledged experiencing symptoms of depression and anxiety. The cluster defined conversational agents, such as chatbots and digital/virtual assistants, as technological tools that facilitate user interactions using natural language. Chatbots equipped with natural language processing can comprehend the patterns in patient conversations and assess the underlying mood of the message by using contextual cues from voice, video or text input (Fitzpatrick *et al.*, 2017; Bendig *et al.*, 2019; Koulouri *et al.*, 2022). A study conducted by Koulouri *et al.* (2022) involved interviews with counsellors who specialize in working with young adults. The

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findings of the study revealed the perceived advantages and potential functions of mental health chatbots as seen by mental health professionals, as well as suggesting specific requirements that need to be met for the technology to be accepted. These research activities have demonstrated that chatbots are a viable solution for providing mental health support to young adults. Furthermore, they advocate for the use of user-centred approaches during the development of mental health chatbots and emphasize the need for more systematic and rigorous evaluations of the resulting solutions. The cluster also illustrates that explainable AI is being integrated into the deployment of digital mental health to get favourable and accountable results. The allocation of resources towards digital platforms and their corresponding applications, which enable immediate screening, monitoring and treatment, has the potential for achieving cost-effectiveness in groups who are particularly susceptible to adverse effects (Balcombe and de Leo, 2020; Balcombe and de Leo, 2021). In regard to future research avenues, the cluster illustrates that it would be beneficial for researchers to create a survey that can collect detailed information on many aspects of mental health and mental health technology (Koulouri *et al.*, 2022). Additional study needs to be conducted to enhance the psychotherapeutic content of chatbots and to assess their effectiveness via clinical trials (Bendig *et al.*, 2019, 2022). Needs assessments might significantly improve the study literature on the creation of chatbots for treatment. Researchers should investigate problems such as what therapists seek along with how chatbots might assist therapists' sense adequately supported by the application (Balcombe and de Leo, 2021; Bendig *et al.*, 2022).

5.4 Cluster 4: Trust and empathy in healthcare

Cluster 4 included articles that examined the significance of trust and empathy in the context of healthcare. The most prominent characteristics in this cluster are stroke prognosis, disease mortality, health care workflows, clinical decision-making, medical informatics [Mesh], dialogue system and technological singularity (Yu *et al.*, 2018; Bohr and Memarzadeh, 2020; Jimma, 2023). The most cited articles in this cluster are Jiang *et al.* (2017), TC = 3117; Laranjo *et al.* (2018), TC = 913; Maruthappu (2018), TC = 393; Ettman and Galea (2023), TC = 319; and Oh *et al.* (2019), TC = 247. The cluster has conducted a thorough examination of the AI applications in stroke, focusing on three key areas: early detection and diagnosis, therapy and outcome prediction and prognosis assessment (Jiang *et al.*, 2017). Moreover, the cluster has defined medical AI as the process of using insights from previous data to inform clinical decision making (Maruthappu, 2018). The cluster discusses assuring the trustworthiness of AI, which entails presuming a well-functioning, rational democracy, whose governing ability will lead the development of AI-based technology (Kraft, 2020). The cluster also discusses the inclusion of essential values-based concerns that highlight factors such as transparency, social justice, informed consent and accountability (Kraft, 2020; Thiebes *et al.*, 2021). Future AI research should prioritize targeted activities that closely fit with current healthcare patterns. Incorporating these systems through clinical practice requires establishing a mutually advantageous partnership between AI and clinicians, in which AI provides clinicians with increased efficiency or cost-effectiveness, and medical professionals provide AI with the necessary clinical exposure for understanding complicated clinical case management (Buch, *et al.*, 2018; Maruthappu, 2018). Potential research should explore applications of chatbots in the psychotherapeutic setting include providing assistance in the areas of preventing, treating and managing psychological issues and mental illnesses, as well as preventing relapses (Laranjo *et al.*, 2018). Researchers should be mindful of the fact that while AI may not replace clinical decision-making, it has the potential to enhance the choices made by doctors (Kretzschmar *et al.*, 2019; Bohr and Memarzadeh, 2020). In resource-constrained environments, AI may be used to perform screening and assessment in the absence of enough medical competence, making it a compelling field of research (Jiang *et al.*, 2017; Davenport and Kalakota, 2019).

6. Theoretical contribution/conceptual model

The present study presents a theoretical framework that investigates the influence of AI characteristics on patient-related outcomes in the context of healthcare. In the past, researchers have conducted a study to explore how empathic behaviour generation in AI chatbots leads to affect recognition, and consequently, explains the conversational agent's behaviour (Castellano *et al.*, 2013). Furthermore, hedonic and pragmatic aspects lead to an effective quality of interaction between the AI and patient, as proposed by the research works of Niculescu *et al.* (2013). Researchers have also asserted that companion's emotional state and perspective-taking increased the empathetic reactions, enabling medical personnel in identifying mental health concerns (Leite *et al.*, 2013b).

Companion's affective state and perspective-taking further strengthened the empathic responses, further assisting the medical professionals identify the mental health concerns (Leite *et al.*, 2013b; Correia *et al.*, 2018). Studies in this field have additionally demonstrated that the supporting actions of empathetic robots result in empathic reactions, which in turn aid in comprehending mental health issues (Leite *et al.*, 2014; Castellano *et al.*, 2017). The use of dialogue initiative and management in the metaphor used by the conversational agent serves to further augment the comprehension of the mental health issues experienced by a patient (Laranjo *et al.*, 2018; Shi *et al.*, 2021). Inclusion of trust, empathy and compassion that are also seen in the metaphors used by conversational agent assist the healthcare professionals to understand mental health concerns (Bærøe *et al.*, 2020a; Kerasidou, 2020). The same has been illustrated in Figure 4.

7. Future research avenues

The present study has methodically identified possible future areas of investigation on the intersection of AI and empathy by categorizing them into four separate groups. Following the PRISMA framework, the article has emphasized several aspects of this field, such as AI agents replicating emotions, adjusting personality styles and systematically dealing with manifestations of uncertainties (Hegel *et al.*, 2006; Riek *et al.*, 2010; Leite *et al.*, 2013a). These domains provide numerous possibilities for academics who are interested in further investigating the incorporation of empathy in AI systems. Prior investigations have shown that the use of AI is becoming more prevalent in online mental health treatments, serving as a complementary addition to conventional mental health support systems (Kretzschmar *et al.*, 2019). Subsequent studies might explore

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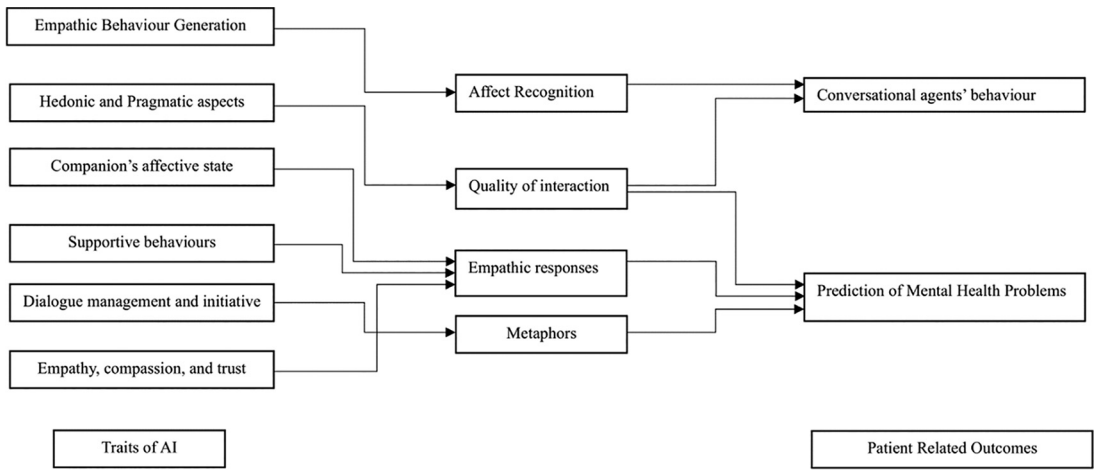
Figure 4 Conceptual model



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the efficacy of these therapies in relation to empathy, assessing their influence on mental health results. Furthermore, the exploration of humanizing robots, which promotes emotional understanding in interactions between humans and machines, is an area of study that shows potential (Giger *et al.*, 2019; Johansson-Pajala *et al.*, 2022). Researchers may investigate the difficulties and limitations of sustaining interpersonal psychological empathy using AI-driven systems. It is of utmost importance to acknowledge and deal with the possible hazards linked to AI (Bærøe *et al.*, 2020b). To ensure the ethical use of AI chatbots as psychiatric tools, it is crucial to do thorough study and use them with precision in clinical settings. This will help to enhance the results of mental health treatment (Vaidyam *et al.*, 2019). Researchers might analyse the incorporation of AI into clinical environments, evaluating its impact on improving performance and delivering healthcare services. Although AI offers potential benefits, its implementation in healthcare systems is presently limited, and there is less data to support its effectiveness in improving procedures or outcomes (Pereira and Díaz, 2019).

Subsequent studies should include direct participation in medical environments to assess the tangible effects of AI on healthcare outcomes. An important obstacle in the field of robotic surgery is the absence of haptic feedback, which refers to the sense of touch (Gumbs *et al.*, 2013; Vaidyam *et al.*, 2019). Research might concentrate on investigating how empathic AI could help reduce this disparity, hence improving the willingness of surgeons to embrace robotic technologies. Furthermore, the importance of accessibility, presentation and transparency in digital tools offers a promising area for further research, especially in developing nations where AI has the potential to enhance business performance while maintaining empathy (Balcombe and de Leo, 2020, 2021).

Future research should also address the inherent biases present in AI systems, with a focus on developing and implementing empathetic AI systems in a fair and equitable manner. Although this work has identified important viewpoints and methods for incorporating empathy into AI, ongoing scrutiny is required to ensure that AI progress is in sync with changing human lifestyles. The clusters found in this study provide several opportunities for further investigation. In order to improve the empathy of social robots, future research should focus on the intricate nature of social behaviours. This may be achieved by boosting the robots' empathy via factors such as voice characteristics, linguistic signals and the ability to recognize and respond to emotions (Castellano *et al.*, 2013; Niculescu *et al.*, 2013). Creating evaluation criteria for empathetic agents and using emotional computing and user modelling are essential for progressing in this area (Yalçın and DiPaola, 2020). Furthermore, it is important for research to examine the genuineness of the perceived robotic assistance in human–robot relationships, as well as the effects of robotic malfunctions on empathy. This has been explored by Castellano *et al.* (2013) and Castellano *et al.* (2017). It is crucial to investigate emotional connections and the ability to control emotions in interactions between humans and robots (Rodrigues *et al.*, 2012; Correia *et al.*, 2018). Furthermore, the impact of conversational agents on mental well-being is being examined. It is essential to improve the psychological treatment aspects of chatbots and carry out clinical experiments to evaluate their efficacy (Bendig *et al.*, 2019, 2022). Adopting user-centered methodologies in the development of mental health chatbots and conducting systematic assessments of these tools will greatly enhance the provision of mental health assistance (Morris *et al.*, 2018; Koulouri *et al.*, 2022).

The fourth topic to be discussed is the importance of trust and empathy in the healthcare field. Ensuring the reliability of AI in healthcare contexts requires the integration of openness, social justice, informed consent and accountability (Kraft, 2020; Thiebes *et al.*, 2021). The use of AI in screening and evaluation, especially in situations with limited resources, continues to be an intriguing field of study (Jiang *et al.*, 2017; Davenport and Kalakota, 2019).

To summarize, this research has made substantial progress in comprehending the significance of empathy in AI, specifically in the healthcare industry. Nevertheless, it is crucial to do more study in order to adjust to technical progress and changing human requirements. This will guarantee that AI continues to be a vital instrument in developing empathy and enhancing

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healthcare results. Subsequent research should further investigate the discovered clusters, using novel perspectives and advancements in technology to cultivate sympathetic AI systems that are in line with the intricate nature of human emotions and healthcare obstacles. Table 5 presents the future study avenues in AI-empathy research, which have been derived from the clusters identified in the literature.

8. Ethical reflections for AI in mental healthcare convention

To use AI ethically, algorithms used to forecast or diagnose mental health diseases must be accurate and not put people at danger. Furthermore, people participating in AI technology selection, testing, implementation and assessment must be conscious of ethical concerns, such as biased information (e.g. subjective and expressive character of clinical text data; linkage of mental diseases to specific ethnicities) (Crigger and Khoury, 2019). Approved ethical concepts guiding biomedical research, such as autonomy, beneficence and justice, must be prioritized and, in certain situations, augmented (Osogami and Raymond, 2019; Jin *et al.*, 2023). It is vital to address data and technological literacy gaps for both patients and professionals. Furthermore, to the best of researcher's knowledge, no clear guidelines exist to guide the use of AI and other developing technologies in healthcare settings (Graham *et al.*, 2019).

Computational researchers may couch AI with the assistance of datasets that dearth sufficient data to make substantial assessments or projections (Haan *et al.*, 2019). Doctors may neither have the knowledge of managing the intensity of granular data, nor be self-assured with a judgement produced by AI (Adibuzzaman *et al.*, 2017; Vaidyam *et al.*, 2019). Institutional Review Boards have a confined information regarding emerging technologies, which causes risk assessment to be incompatible and unpredictable (Graham *et al.*, 2019; Sussman *et al.*, 2022).

For instance, there are endeavours to connect smartphone keystrokes and voice prototypes to mood, but then people in general may not know such linkages are possible (Tsoi *et al.*, 2021). Public correspondence about these calculations should be valuable, logical and present that tools supplement, however, do not substitute, clinical practice. Therefore, there exists a need to coordinate ethics into the advancement of AI by means of examination and training and assets should be appropriated for this reason.

Table 5 Future research directions in AI-empathy research based on the clusters

Clusters	Exemplary questions
Social robots and empathy	• What is the significance of empathy in enhancing performance of a robot-assisted therapy? • To what scale do empathic emotions influence the functioning of social robots? • What is an essential component for the future of computational empathy to leverage cutting-edge research in emotional computing and user modelling? • What are the unique assessment metrics and surveys validated specifically for empathetic agent research? • What additional aspects, other than empathy generation, affect recognition and emotional AI-driven sentiment analysis that may be recognized as key components in boosting empathy in social robots?
Empathy in human–robot relations	• How does AI assist in meeting the shortage of medical practitioners for individuals dealing with mental disorders? • How can emotional signals and empathic sentiments be explored, which cultivate through the modification of affective interactions, mood, and personality? • How does AI assist in comprehending and illustrate signs of clinical depression, such as poor vocabulary, semantic incoherence, and an absence of syntactic complexity? • How can the repercussions of an unexpected malfunction or the robot's inability to display appropriate supportive behaviour in response to a possibly minor situation be examined? • How does the empathy driven AI operate in developing nations like India, Brazil among others?
Conversational agents and mental well-being	• What research can be conducted to improve the psychotherapy content of chatbots and test their usefulness via clinical trials? • How can research address what therapists desire, as well as how chatbots may aid therapists in feeling appropriately supported by the application? • What are the possible attributes for a conversational agents/chatbot? • What needs evaluations may be carried out to substantially improve the research literature on the development of chatbots for therapy? • How can a survey be constructed that can gather extensive information on several elements of mental health and mental health technology?
Trust and empathy in healthcare	• What is the role of empathy in enhancing the presence of AI in healthcare sector? • How can research be prioritized to build trust in AI by focusing on activities that resemble current healthcare patterns? • How can applications of chatbots be investigated in the psychotherapeutic context, especially in the areas of preventing, treating, and managing psychological disorders and mental diseases? • How can AI be used in resource-constrained contexts to undertake screening and evaluation in the lack of sufficient medical knowledge, making it a promising subject of study? • How can researchers include fundamental values-based considerations, such as transparency, social justice, and accountability, into their potential studies, that eventually promotes trust in AI?

Source: Table by author

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9. Conclusion

The present study makes a substantial contribution by identifying the pertinent ideas, settings, and factors related to empathy in the field of AI literature. Moreover, it sheds light on potential avenues for future research in this subject. This research highlights the importance of empathy in AI, specifically in the healthcare industry, with the goal of enhancing existing capabilities to more effectively address the requirements of healthcare consumers. This study examines the impact of empathy in the healthcare industry and identifies areas where improvements are needed. By doing so, it connects theoretical concepts with real-world application.

Like every research, this study has certain limitations. The article search is limited to the Scopus and WOS databases. To conduct a more thorough study, future researchers should consider expanding their research to include other databases such as Google Scholar, JSTOR, ProQuest or other relevant sources. Subsequently, researchers could review these findings from several databases to strengthen the reliability of the conclusions made in this study. Furthermore, this study does not include grey literature, such as doctorate theses, conference papers, seminar presentations and other sources that have not undergone peer review. Future research endeavours could strengthen the existing results by using grey literature, which may provide significant insights that are not included in conventional academic databases. Moreover, it would be beneficial for future studies to investigate the viewpoints of medical professionals in both advanced economies, where AI has been actively implemented in healthcare contexts and developing nations, where the integration of AI is still in its early stages. Gaining insights into the perspectives of healthcare professionals on the incorporation of AI with empathy might provide significant contextual information for the findings of this research. Their perspectives might have a crucial impact in addressing the complicated aspects and limitations linked to AI, while fostering its potential to substantially enhance the healthcare sector.

9.1 Implications for research

The results of this study emphasize important areas that need more research. Subsequent investigations might broaden their focus by including a wider array of databases beyond Scopus and WOS, encompassing research publications, conference papers and pertinent information from other sources. In addition, the incorporation of grey literature might provide a more thorough understanding. Conducting comparative research that gathers the perspectives of medical practitioners from both developed and developing nations might improve our understanding of how to deploy AI with empathy. This will contribute to the existing body of knowledge.

9.2 Implications for practice

The practical ramifications of this study are clearly apparent in its capacity to impact healthcare practices. Through the comprehension and incorporation of empathy in AI, healthcare practitioners have the potential to enhance patient care and satisfaction. Involving medical professionals from both established and developing areas will provide useful perspectives on the practical obstacles and advantages of deploying empathic AI, thereby improving the effectiveness and acceptability of AI in the healthcare sector.

9.3 Implications for society

This study has a substantial social influence as it shapes public opinions about the incorporation of AI in healthcare. The article seeks to enhance the quality of life for patients and make healthcare more responsive and compassionate by focusing on incorporating empathy into AI. The results may provide guidance for public policy, promoting the use of sympathetic AI solutions in healthcare environments to successfully satisfy the needs of healthcare customers.

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